

Python Installation & Virtual Environment Setup (Windows & macOS)

1. Install Python

Windows

- Download Python from <https://www.python.org/downloads/>
- Run the installer.
- Check **Add Python to PATH** and **Install launcher for all users**.
- Verify: `python --version` or `py --version`

macOS

- Download Python from python.org and install.
- Verify: `python3 --version`

2. Install Visual Studio Code

Download VS Code from <https://code.visualstudio.com/>
Install the **Python** extension by Microsoft.

3. Create a Project Folder

Create a folder (for example AI-Class) and open it in VS Code.

4. Open the Terminal

VS Code → **Terminal** → **New Terminal**

5. Create a Virtual Environment

Windows

```
python -m venv .venv  
or py -m venv .venv
```

macOS

```
python3 -m venv .venv
```

6. Activate the Virtual Environment

Windows (PowerShell)

```
.\.venv\Scripts\Activate.ps1
```

If you get 'running scripts is disabled':

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser  
or  
Set-ExecutionPolicy Bypass -Scope Process
```

Windows (CMD)

```
.venv\Scripts\activate.bat
```

macOS / Linux

```
source .venv/bin/activate
```

7. Verify Activation

Your prompt should begin with (.venv).

8. Install Packages

```
pip install openai  
pip install pandas  
View installed packages:  
pip list
```

9. Deactivate

```
deactivate
```

Common Issues

'python' is not recognized

Ensure Python is installed and added to PATH.

Running scripts is disabled

Run:

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

python3: command not found

Reinstall Python or verify installation.